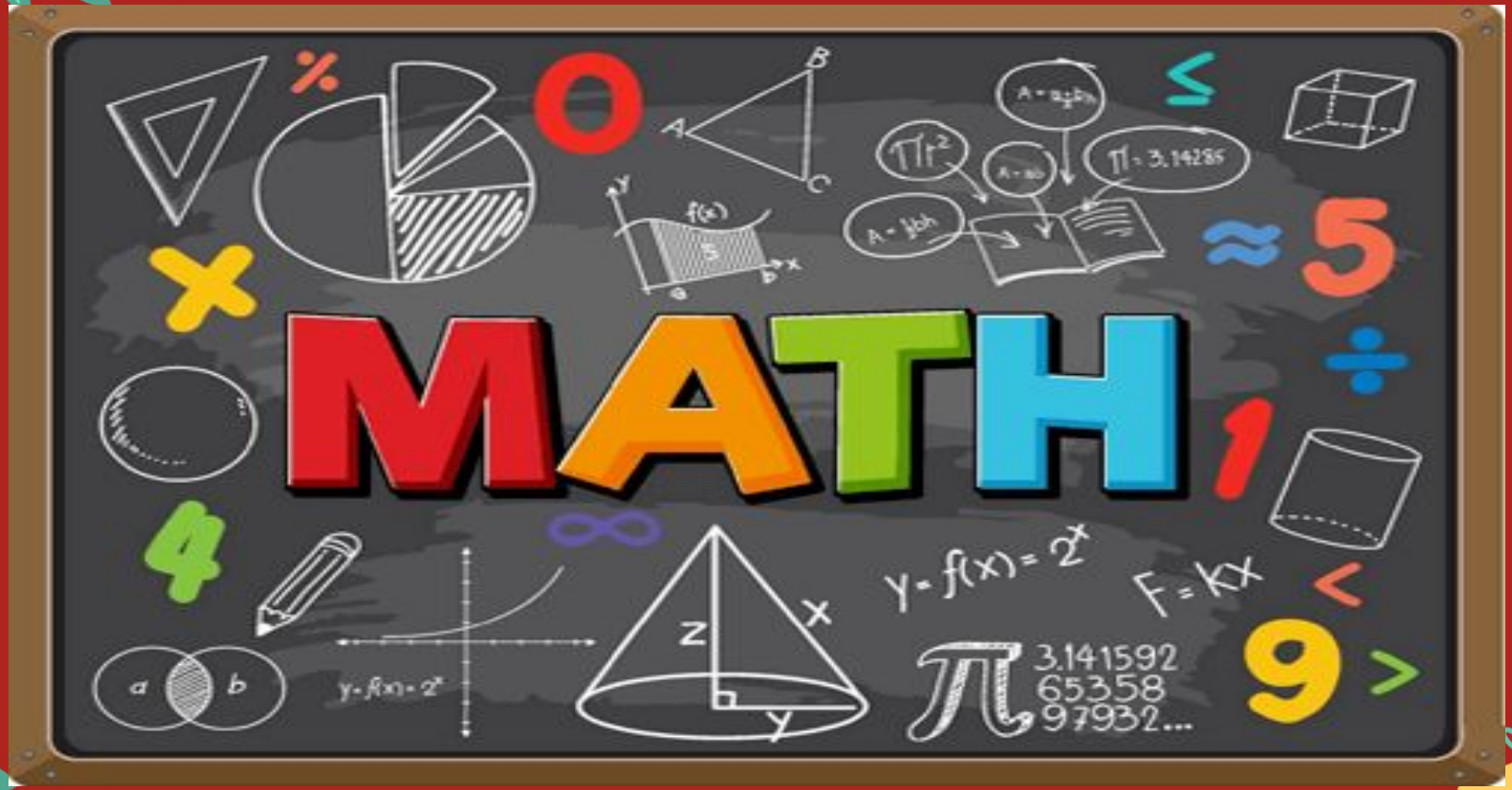

MATH
y = f(x) = 2^x
F = KX
3.141592
65358
97932...

# On This Day In History

On December 3, 1910, a French engineer named Georges Claude showed modern neon lighting to the public for the first time at the Paris Motor Show. He used long glass tubes filled with neon gas, and when electricity passed through the gas, the tubes glowed a bright red-orange color, which amazed people and later led to the neon advertising signs seen in cities around the world.

# Minds-On

- When I say the word "angle" what comes to mind?

# Learning Goal

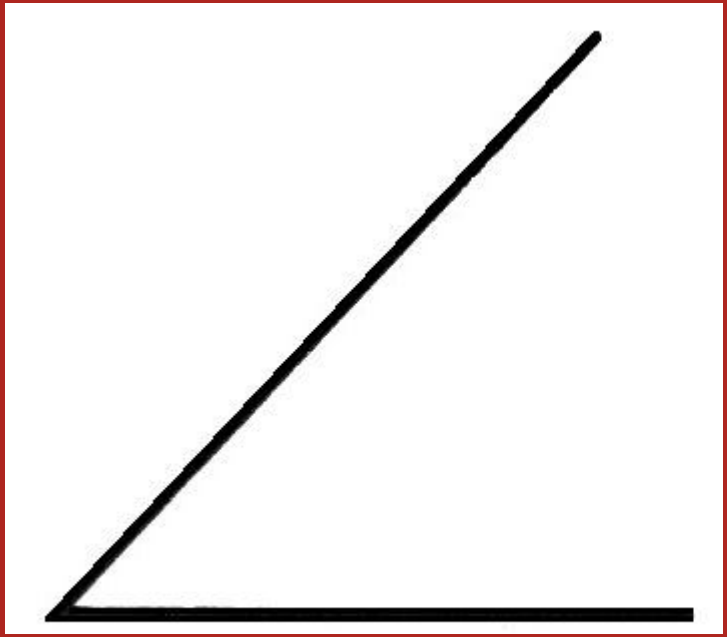
- We are learning how to look at angles and classify them based on their characteristics.

# Success Criteria

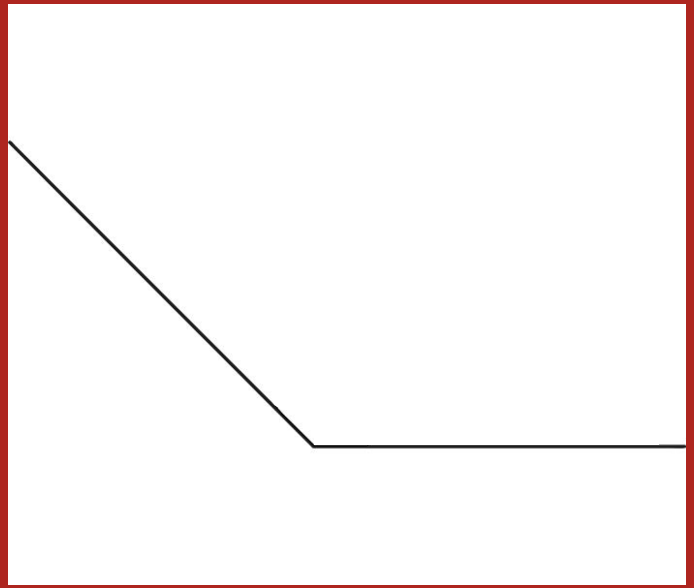
I can...

- Describe what I notice about different angles.
- Explain at least one way angles can be different from each other.
- Sort or discuss angles using the ideas and language from today.
- Work respectfully and responsibly with my group during the activity.

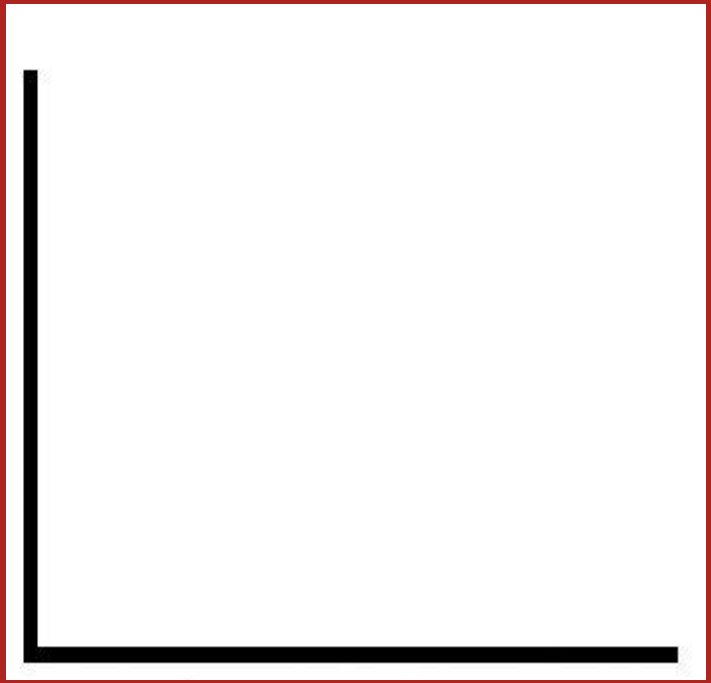
# Angle #1



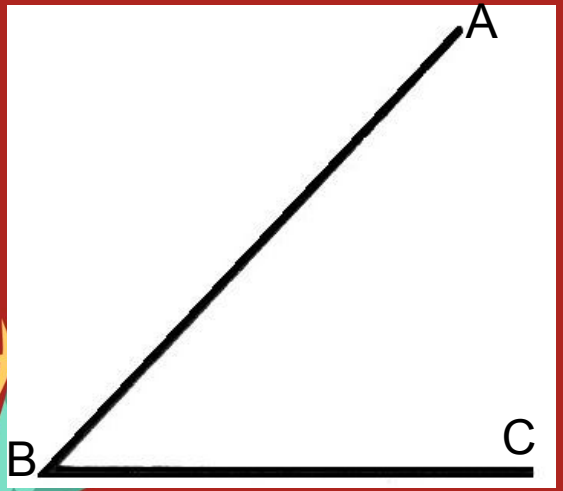
# Angle #2

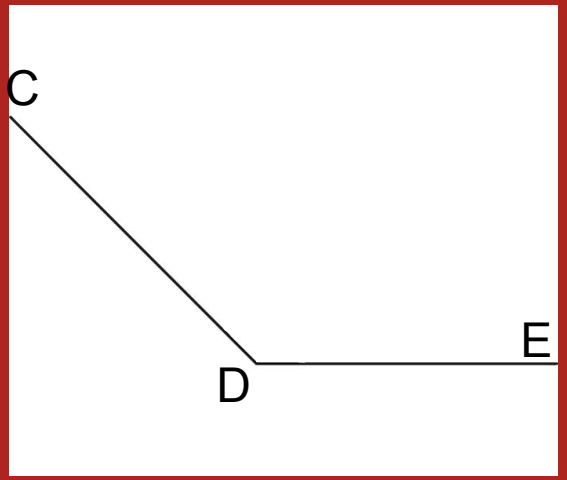


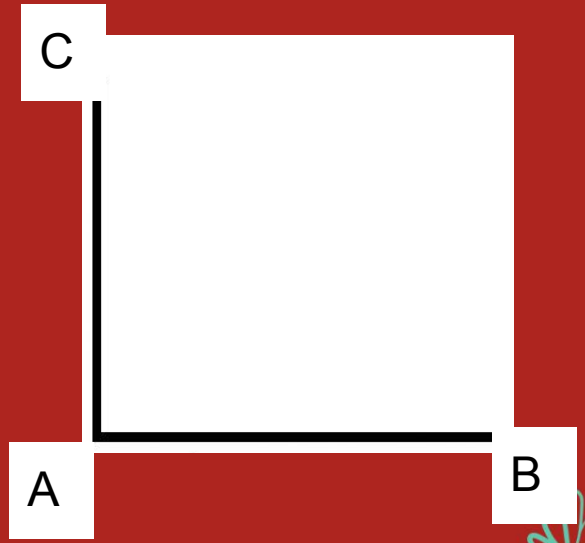
# Angle #3



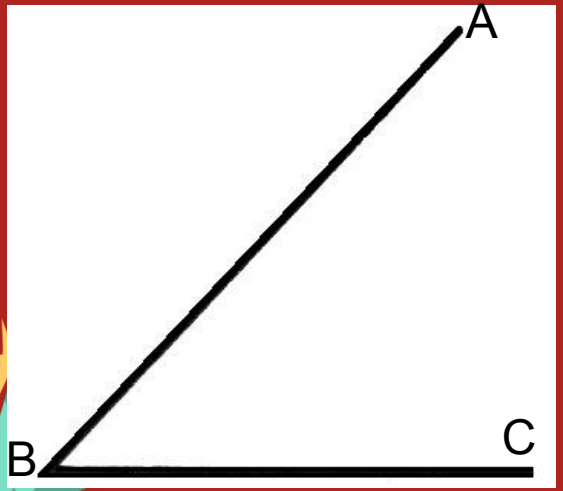
# Adding Labels...

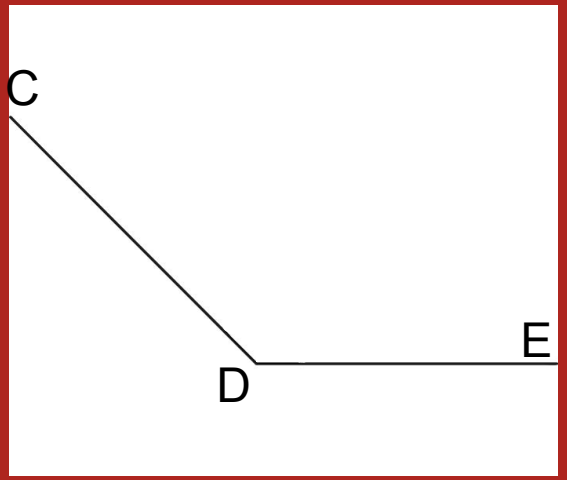


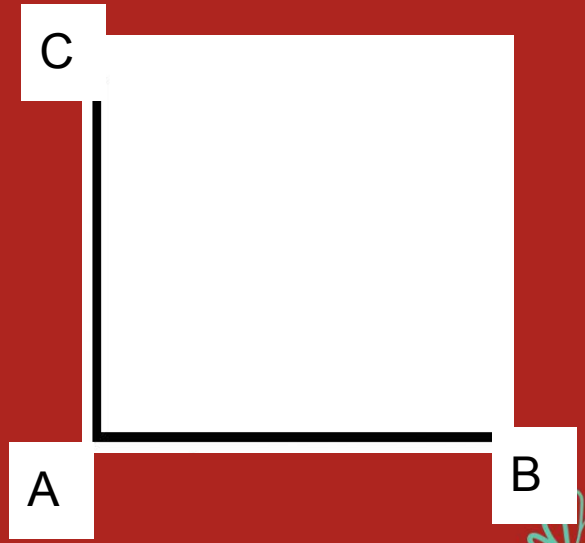



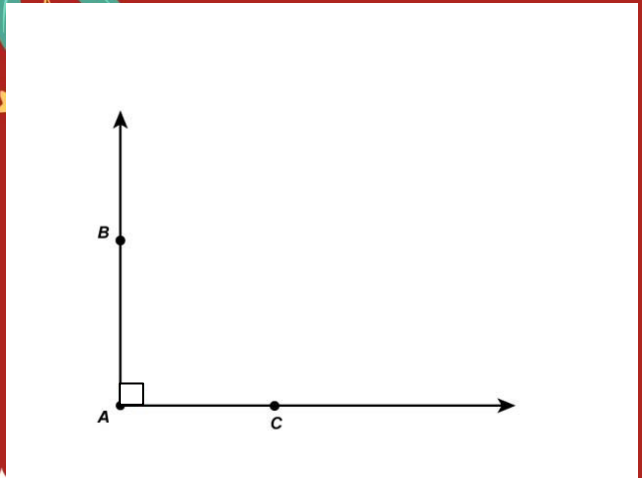
# Adding Labels...

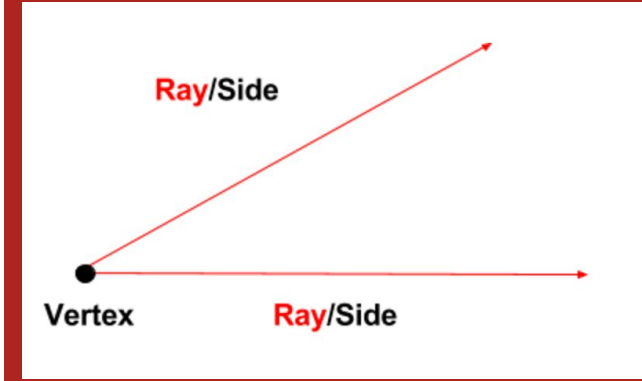


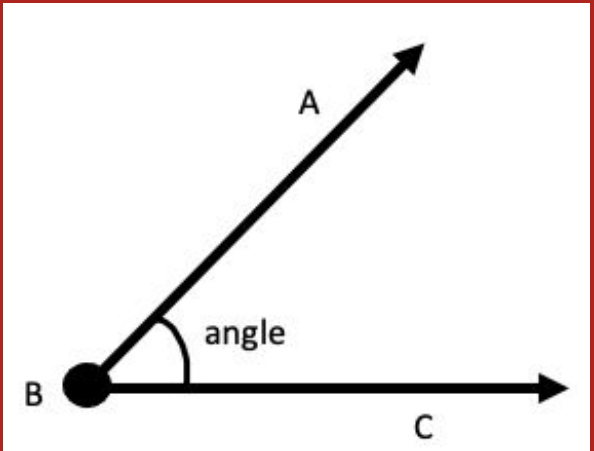



# Symbols for angles



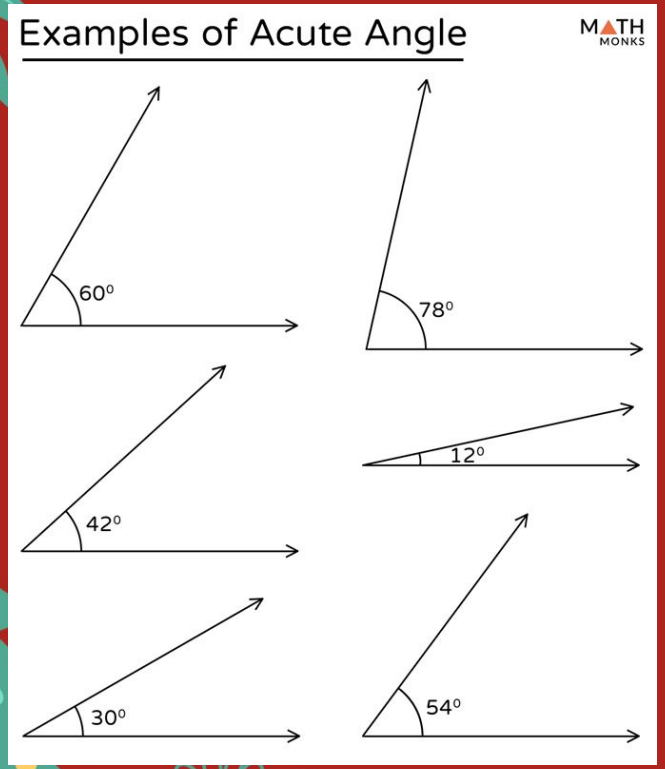



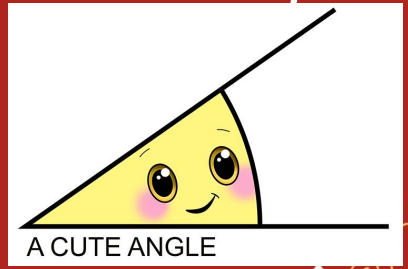

∠BAC

∠ABC

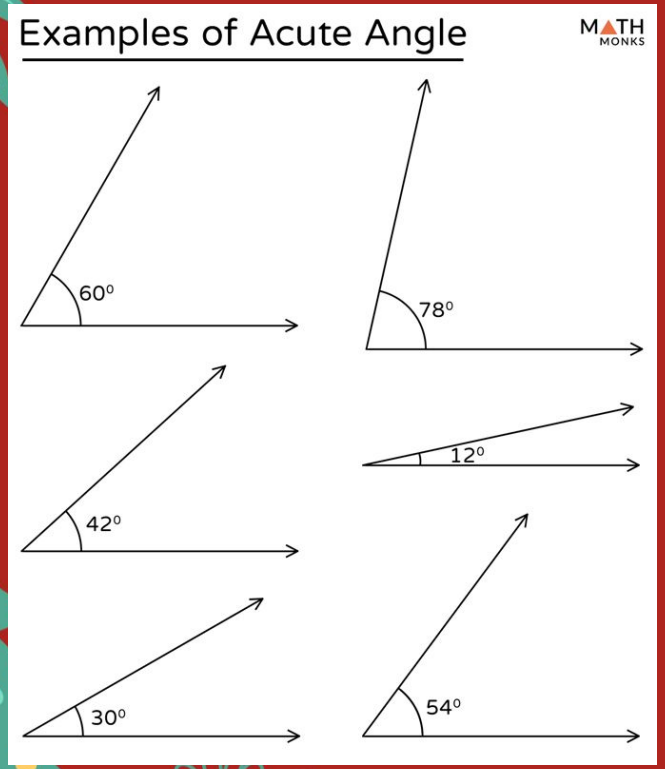
# ACUTE Angle




- An acute angle is an angle that looks <u>small</u> or <u>narrow</u>
- The two <u>rays (a.k.a side)</u> are closer together and the opening is tight.
- "ACUTE" is tiny and cute.



# ACUTE Angle

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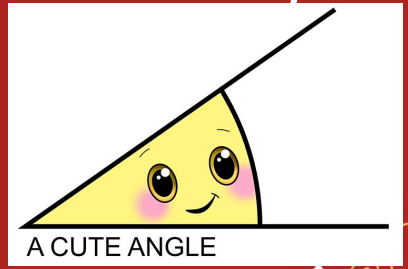
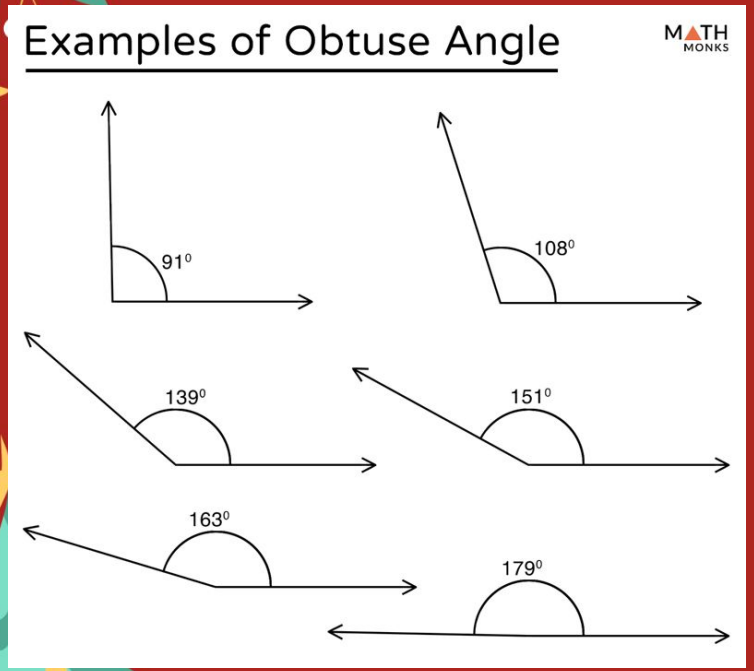
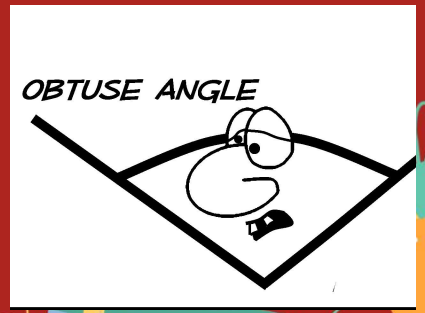


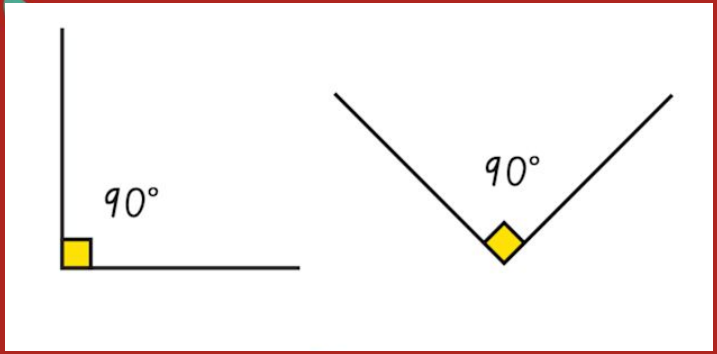
# OBTUSE Angle

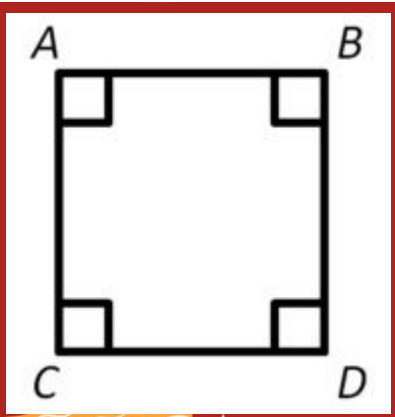



- An obtuse angle is an angle that looks <u>wide</u> or <u>open.</u>
- The two <u>rays (a.k.a side)</u> are spread out more, creating a bigger opening.
- It <u>DOES NOT</u> form a sharp corner.
- "OBTUSE" is big.



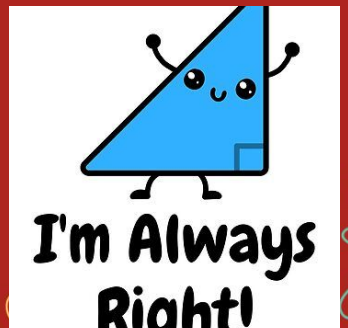
# Right Angle



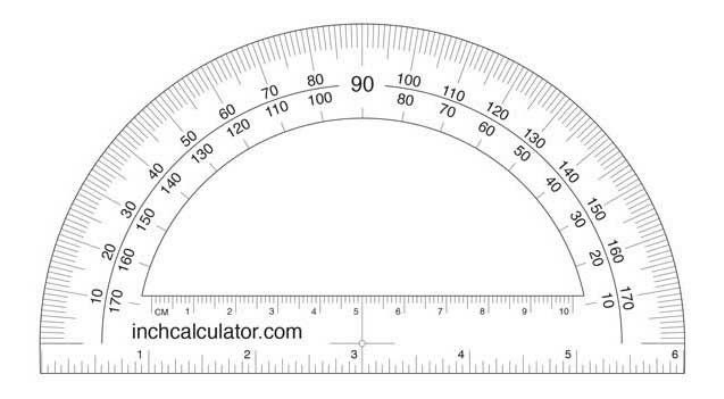

- A right angle makes a <u>perfect square corner</u> (90 degrees).
- The two <u>rays (a.k.a side)</u> are forming a letter L.
- There is a square at the vertex (corner) instead of a circle.
- "Right" is like

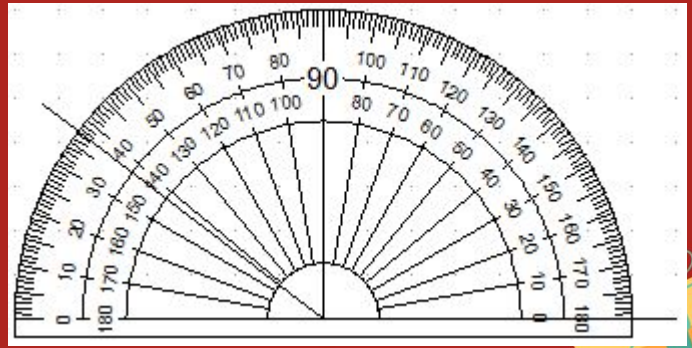
the letter L.



# How Do We Measure Angles?

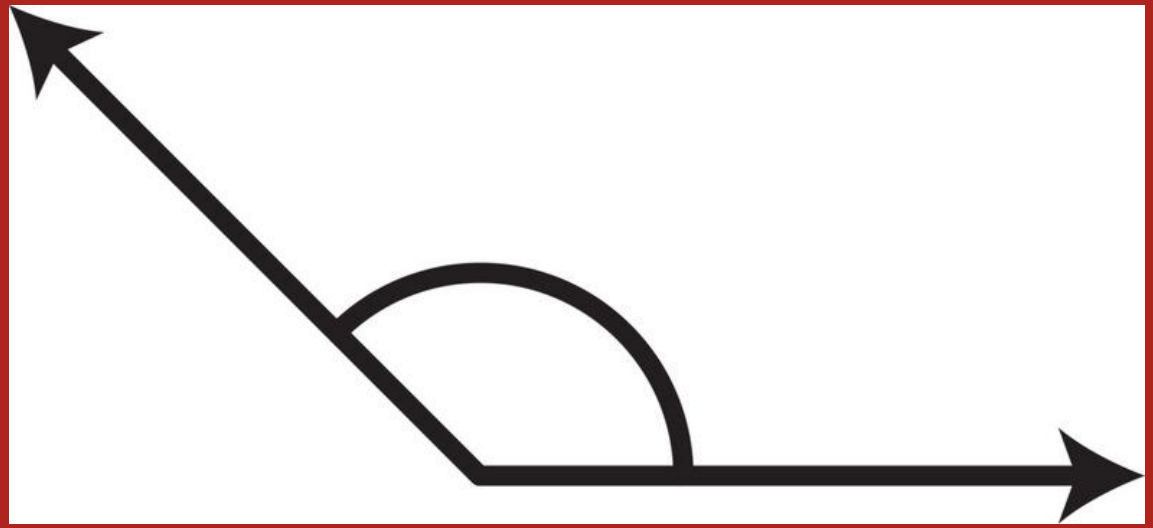
- A protractor is a tool that helps us measure how wide an angle opens. It is important because it lets us describe angles and compare them in a precise way.



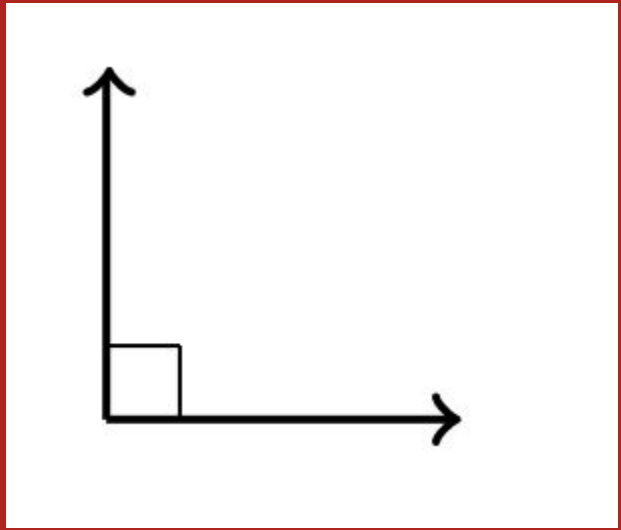
Thumbs Up (Yes) or Down (No)

Is this an OBTUSE angle?



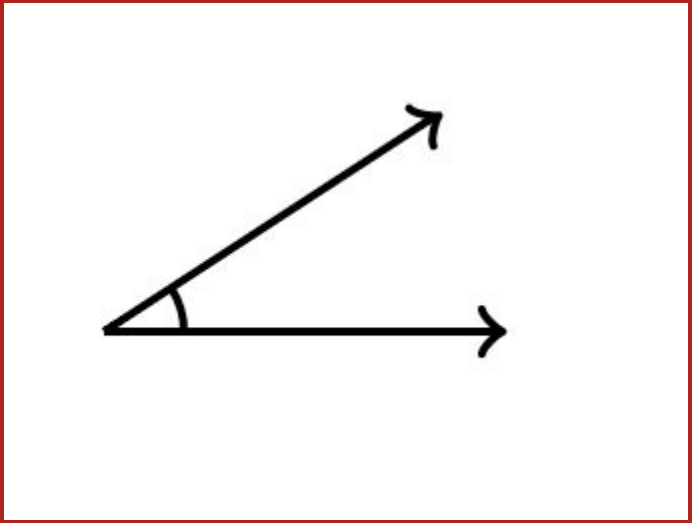
Thumbs Up (Yes) or Down (No)

Is this an OBTUSE angle?



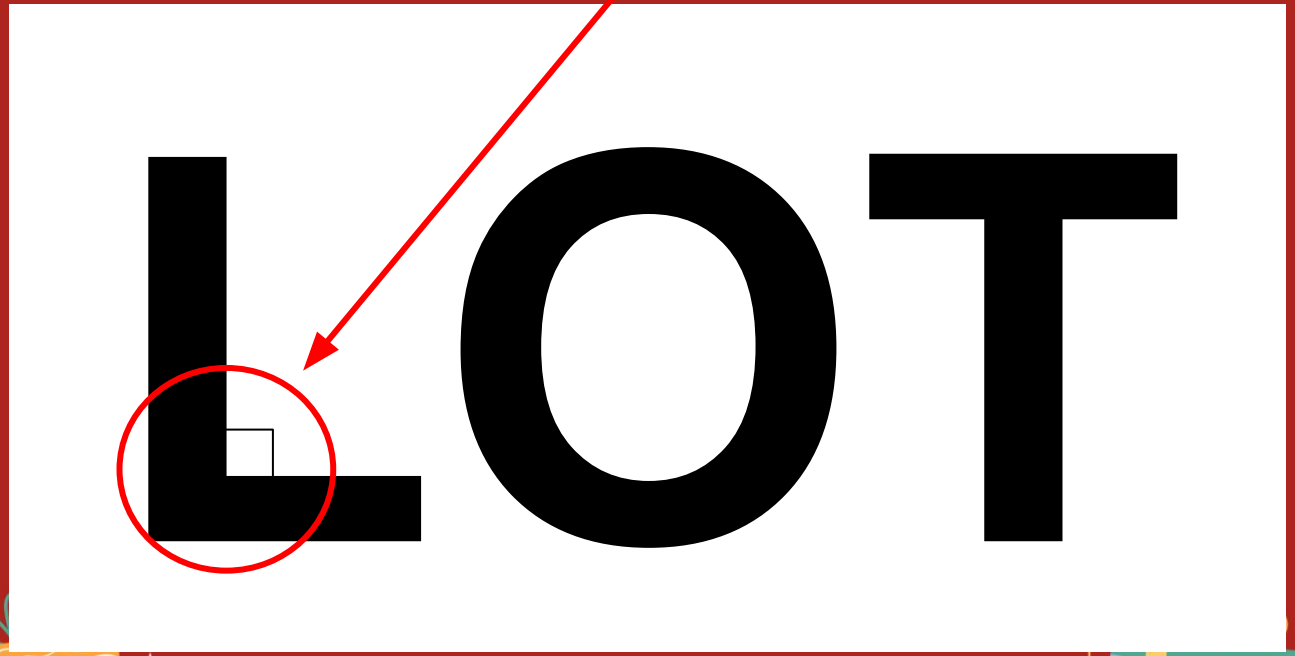
# Thumbs Up (Yes) or Down (No)

## Is this an ACUTE angle?



Thumbs Up (Yes) or Down (No)

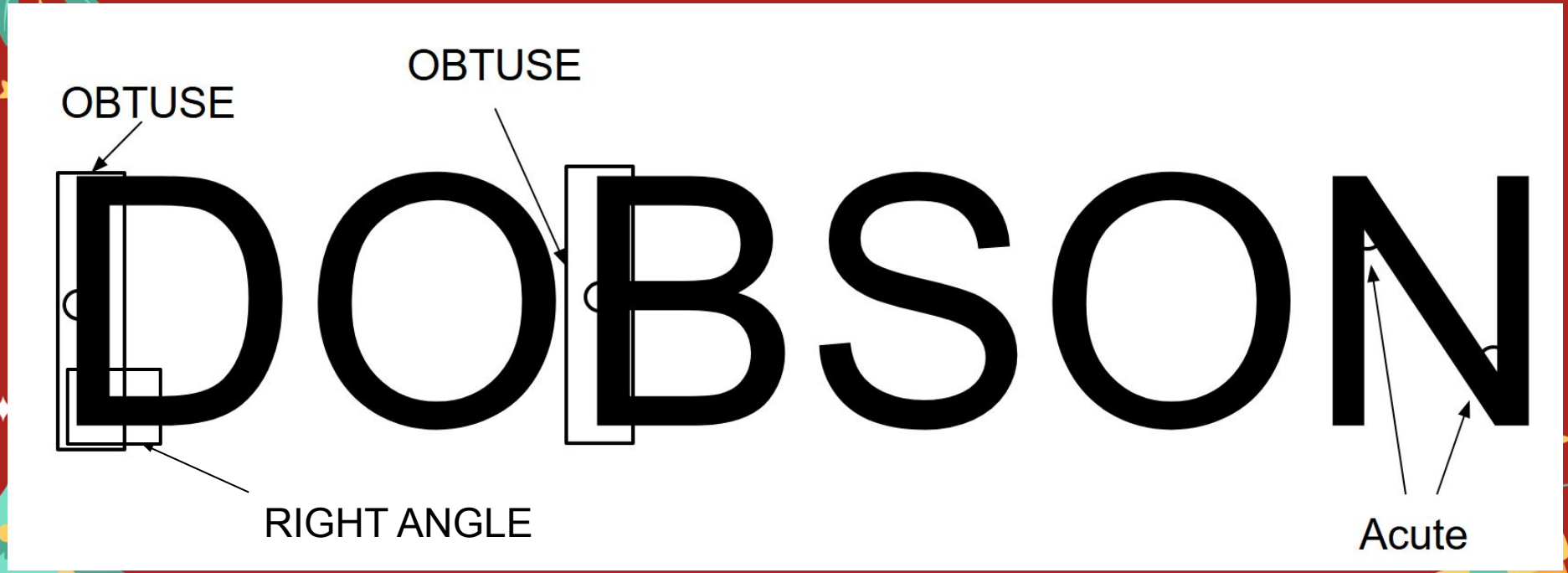
Is this a RIGHT angle?

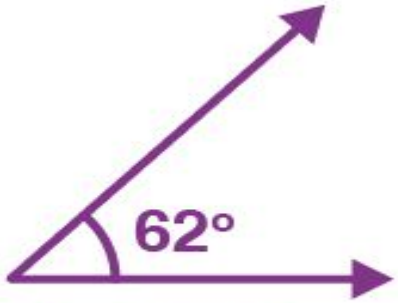


# Activity Outline

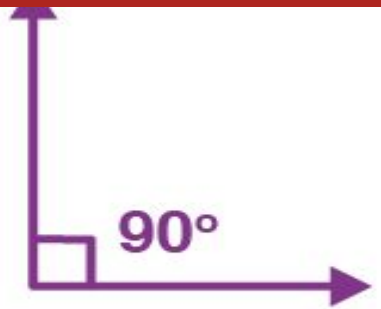
- In your assigned groups, you are going to work on the worksheets provided.
- Each worksheet has YOUR name on it. You are to find the different angles we learned about today and label them on your worksheet.
- If you finish early, turn your sheet over and look through the letters of the alphabet. See how many different angles you can find and label on the back

# Example of Activity - Mr. Dobson






Acute angle




Right angle

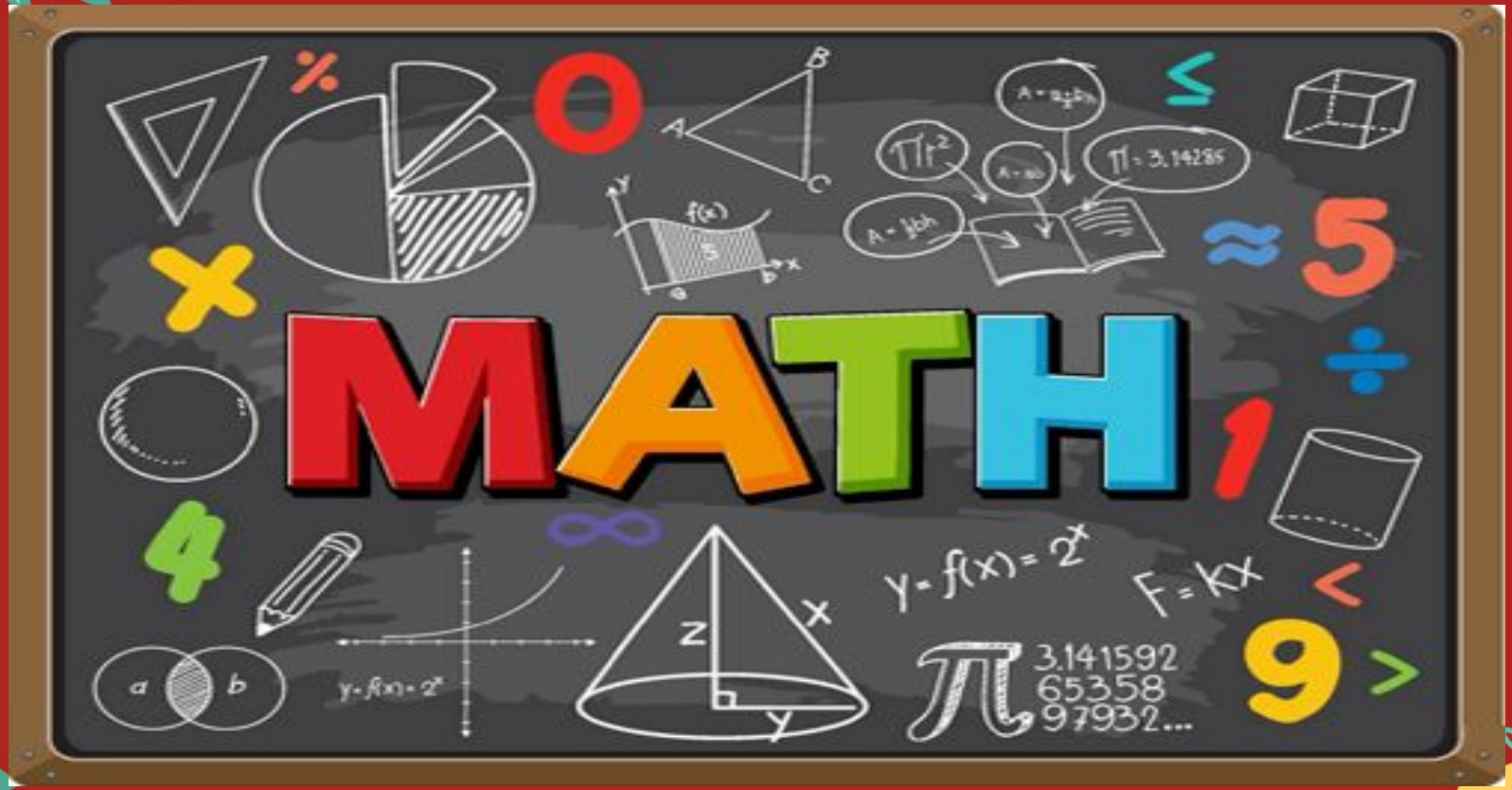
Obtuse angle

| | | |
|---|---|---|
| • Has a small opening<br>• Looks narrow or sharp<br>• Lines are closer together | • "Square like corner"<br>• Makes a perfect corner<br>• Marked with a small square | • Looks open or spread out<br>• Larger opening than an acute angle |

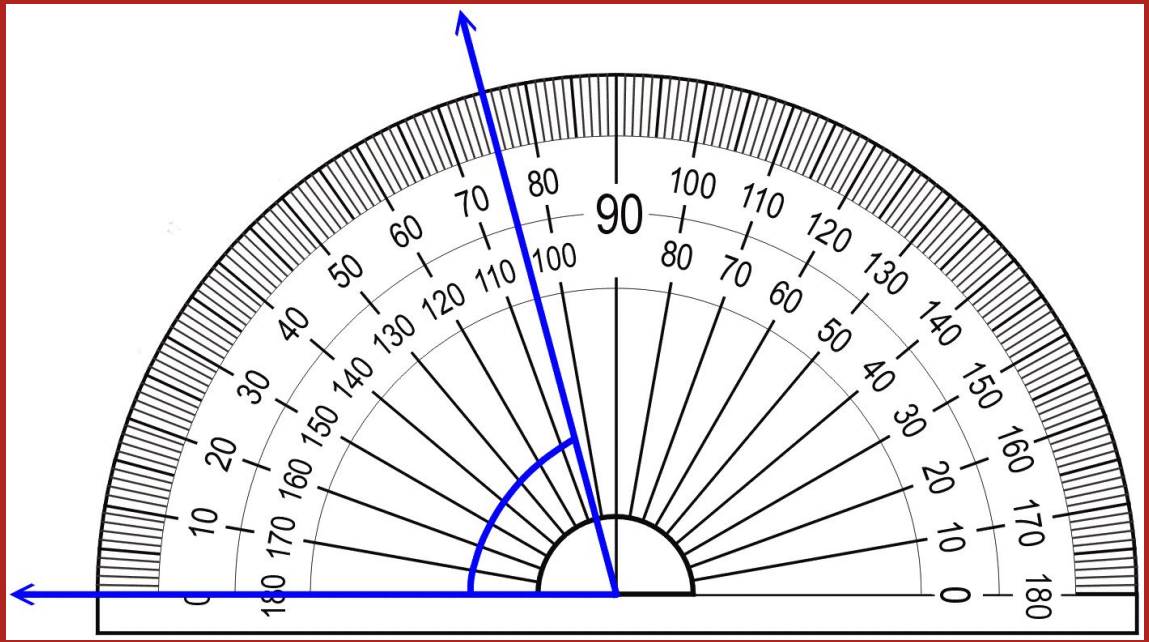
# Success Criteria

I can...

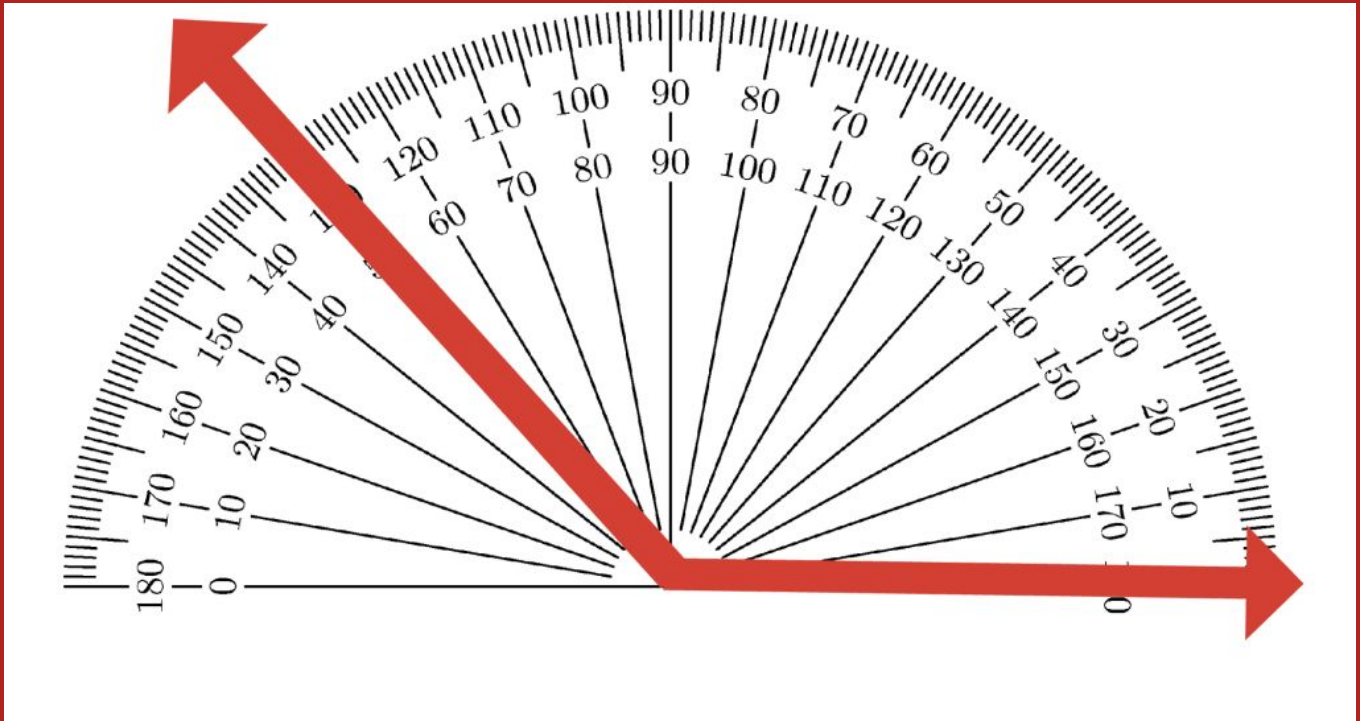
- Describe what I notice about different angles.
- Explain at least one way angles can be different from each other.
- Sort or discuss angles using the ideas and language from today.
- Work respectfully and responsibly with my group during the activity.


MATH
y = f(x) = 2^x
F = KX
3.141592
65358
97932...

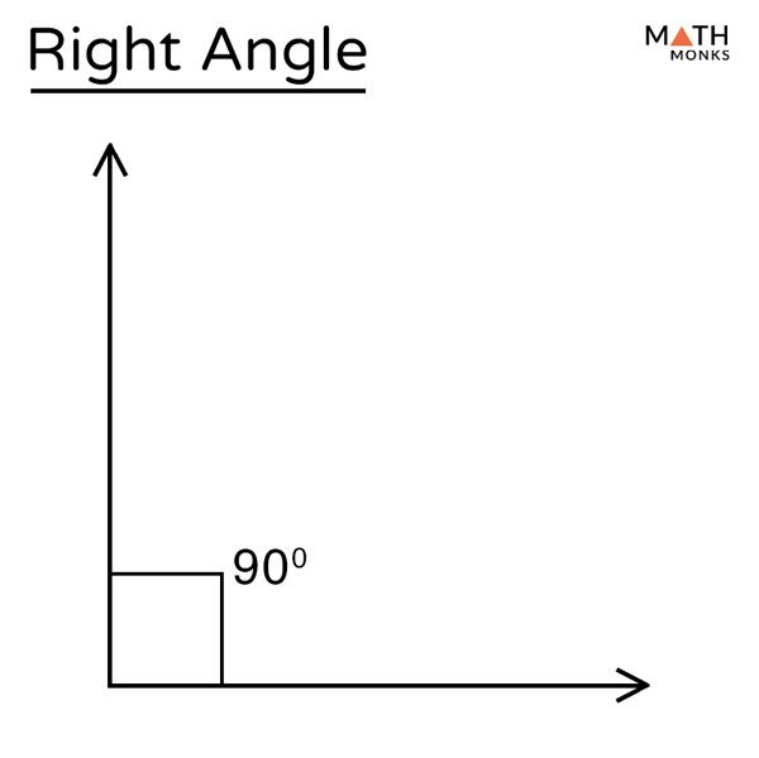
# Measuring Example 1



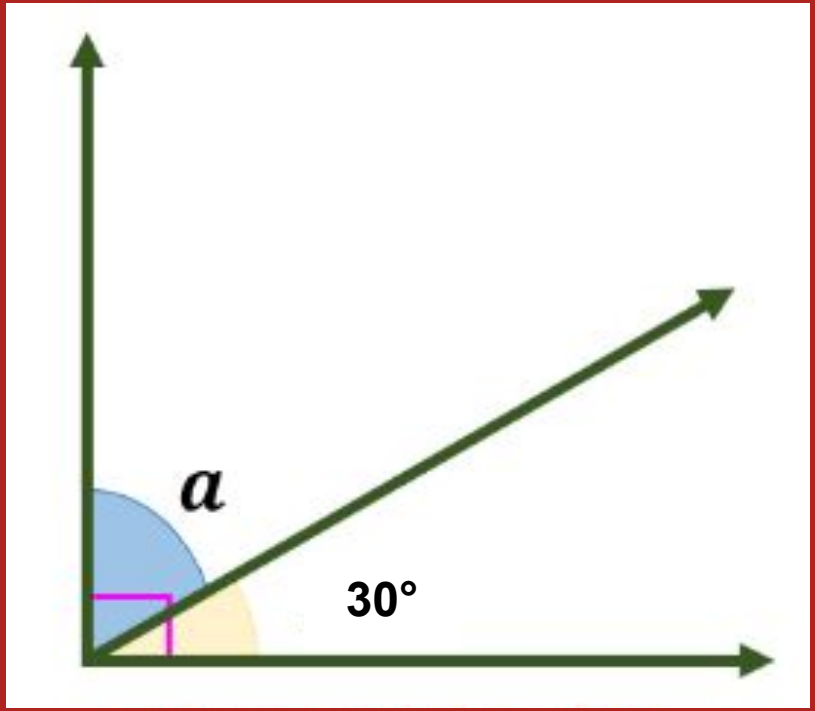
# Measuring Example 2



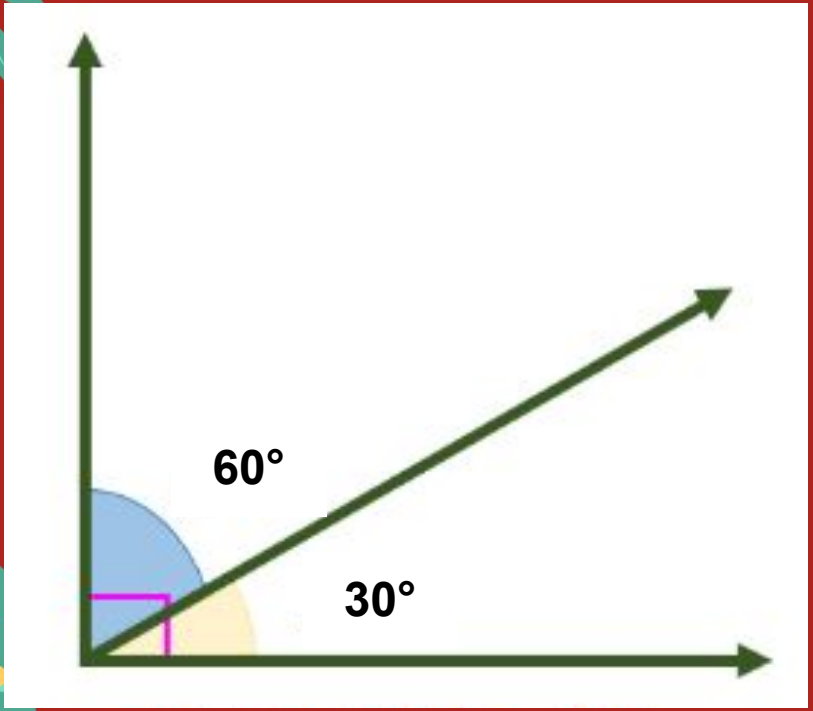
# Right Angle



# What Happens When We Split a Right Angle?



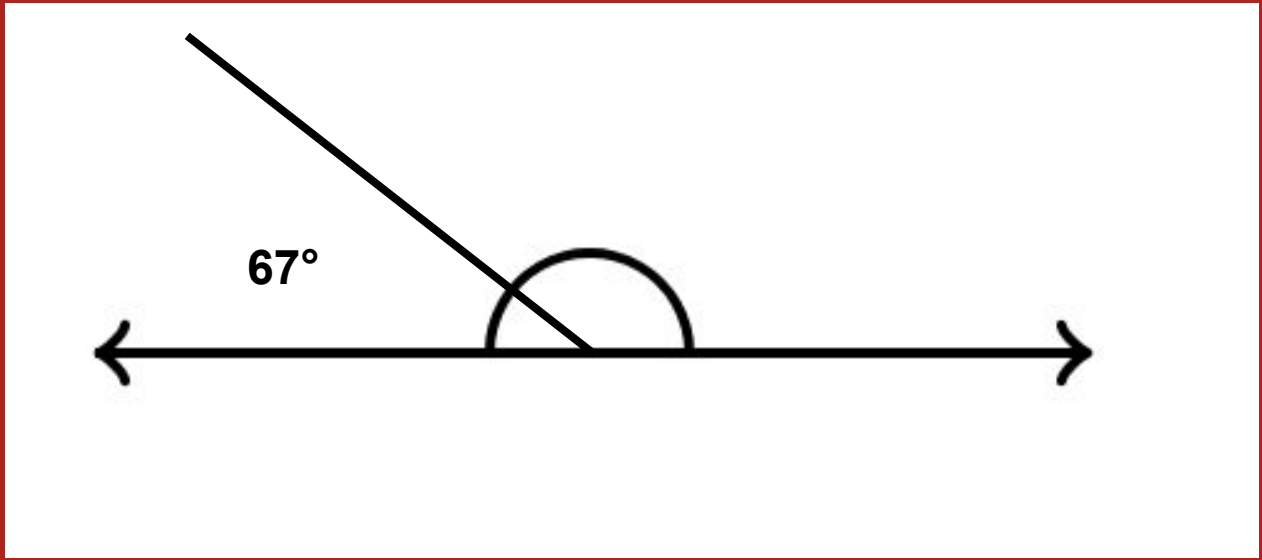
# What Are Complementary Angles?



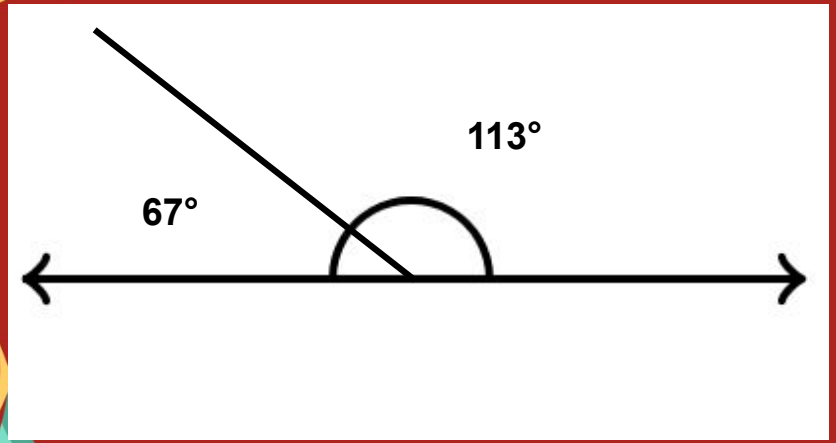

- These two angles add up to 90°, they form a Right Angle together!
- If one angle changes, what would happen to the other angle?

# Straight Angle

# What Happens When We Split a 180° Angle?



# What Are Supplementary Angles?




- Supplementary angles are two angles that add up to 180°.
- They often appear on a straight line, but do not need to be touching.
- Any time two angles total 180°, they are supplementary.

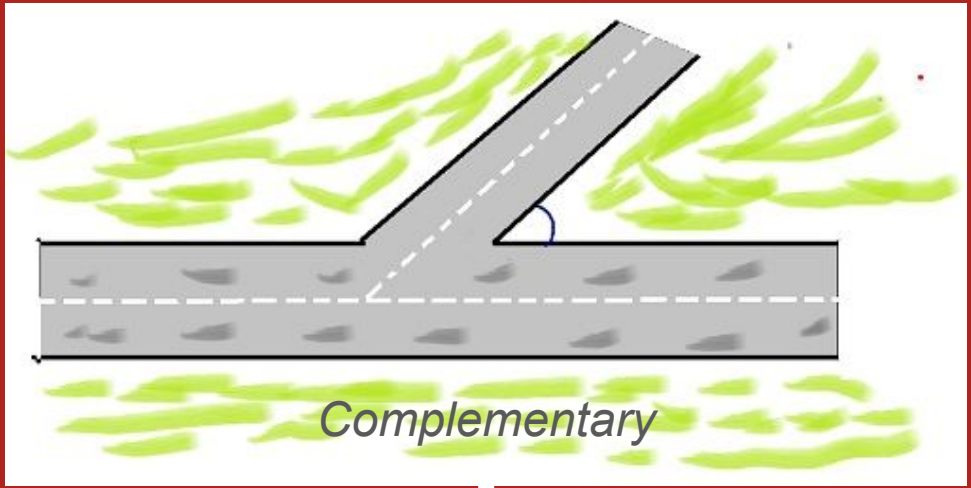
# Complementary or Supplementary?

Angle A = 35° and Angle B = 55°

Complementary?

Supplementary?

# Complementary or Supplementary?




Complementary?

Supplementary?

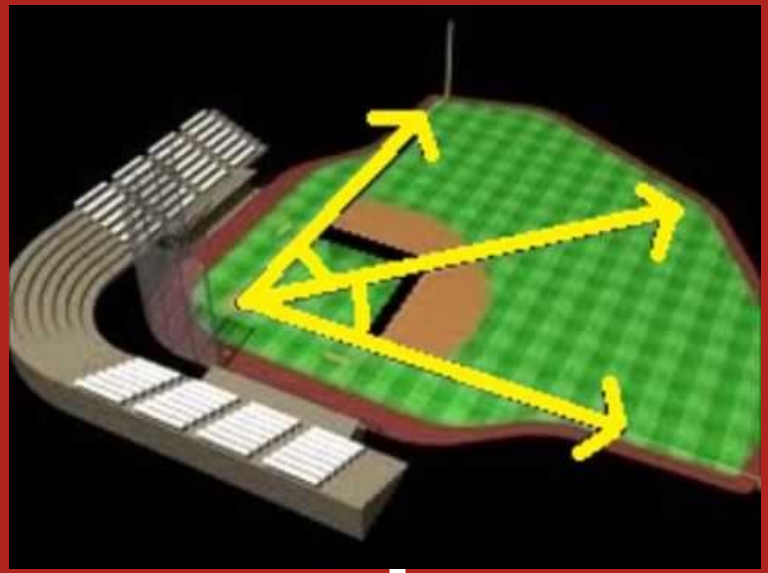
# Complementary or Supplementary?

Angle A = 135° and Angle B = 45°

Complementary?

Supplementary?

# Complementary or Supplementary?



Complementary?

Supplementary?

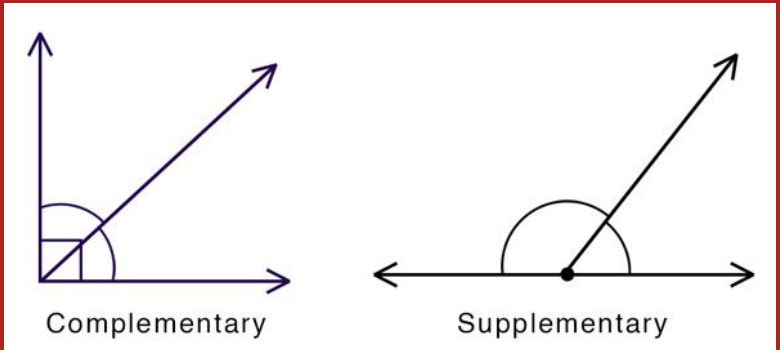
# Complementary or Supplementary?

Angle A = 35° and Angle B = 55°

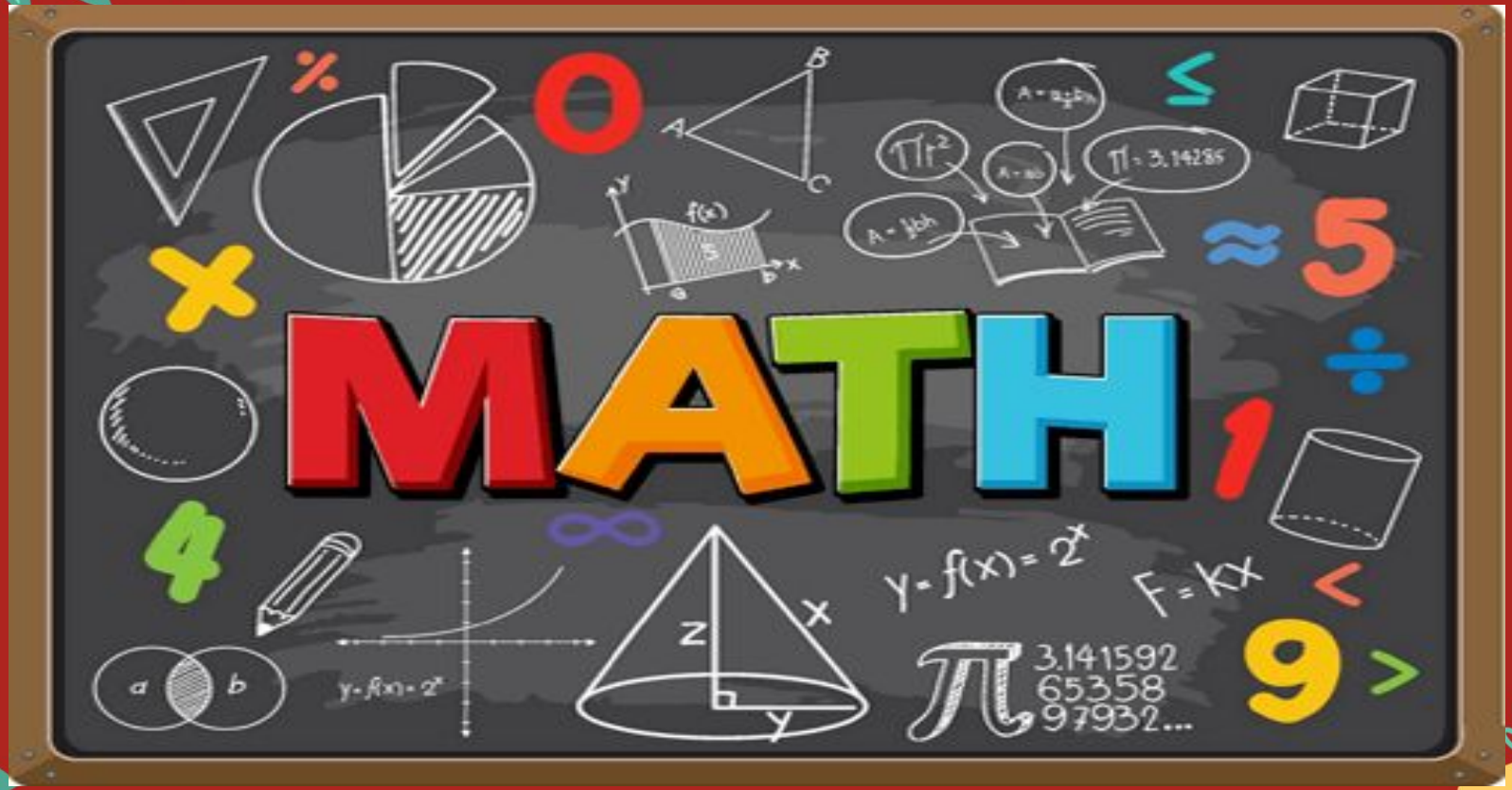
Complementary?

Supplementary?

# Complementary vs Supplementary Angles




| Complementary Angles | Supplementary Angles |
|---|---|
| Add up to **90°** | Add up to **180°** |
| Form a **right angle** | Form a **straight angle** |
| Example: **30° + 60°** | Example: **120° + 60°** |
| If one angle gets bigger, the other must get smaller so the total stays 90°. | If one angle gets bigger, the other must get smaller so the total stays 180°. |


MATH
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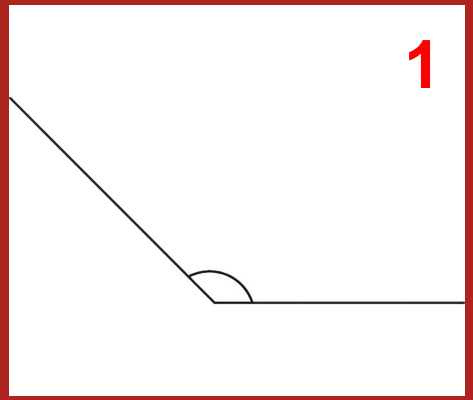
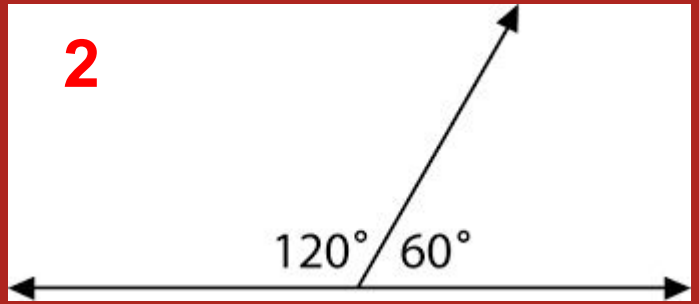
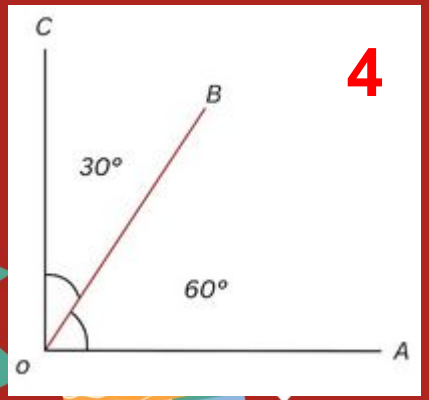
97932...

# Warm Up



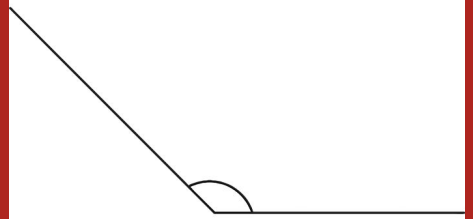






Angle a is 130° and angle b is 50°

5

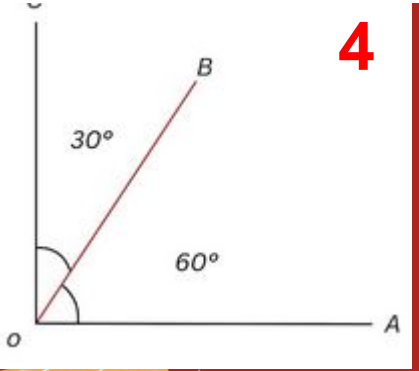
1



Obtuse

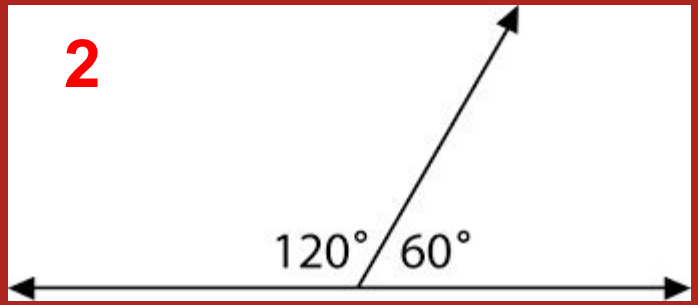
Complementary Angles

4




Supplementary Angles

2




Supplementary Angles

3

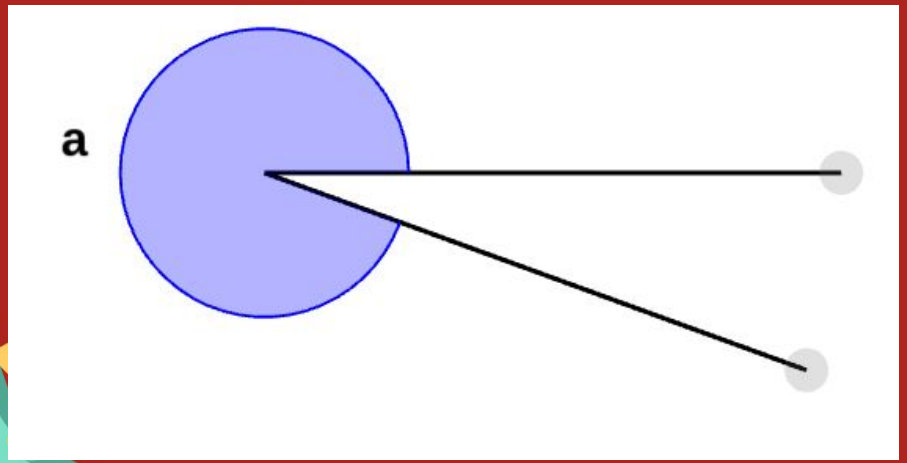
Acute

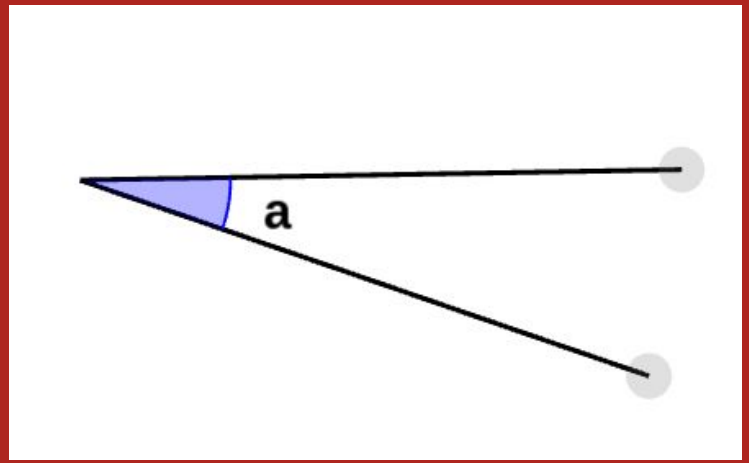
Angle a is 130° and angle b is 50°

5

# Warm Up - Bonus

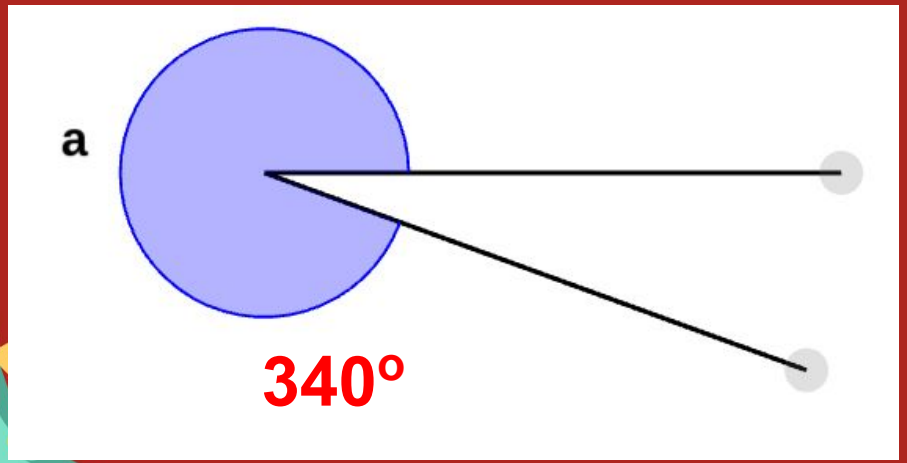
Are these two angles the same?

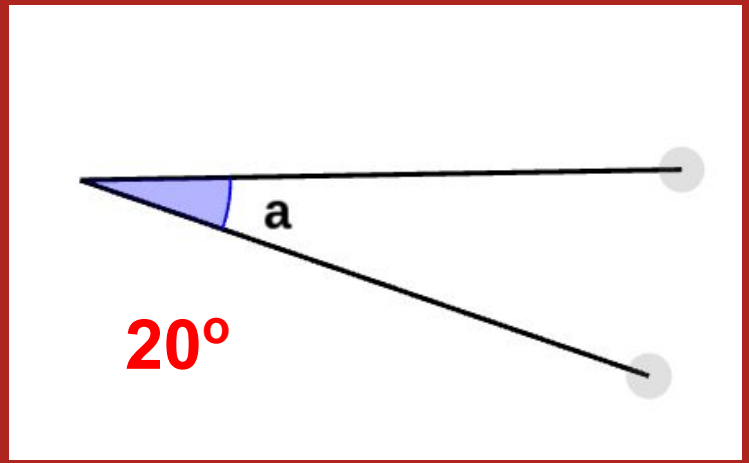


# Warm Up - Bonus

Are these two angles the same?

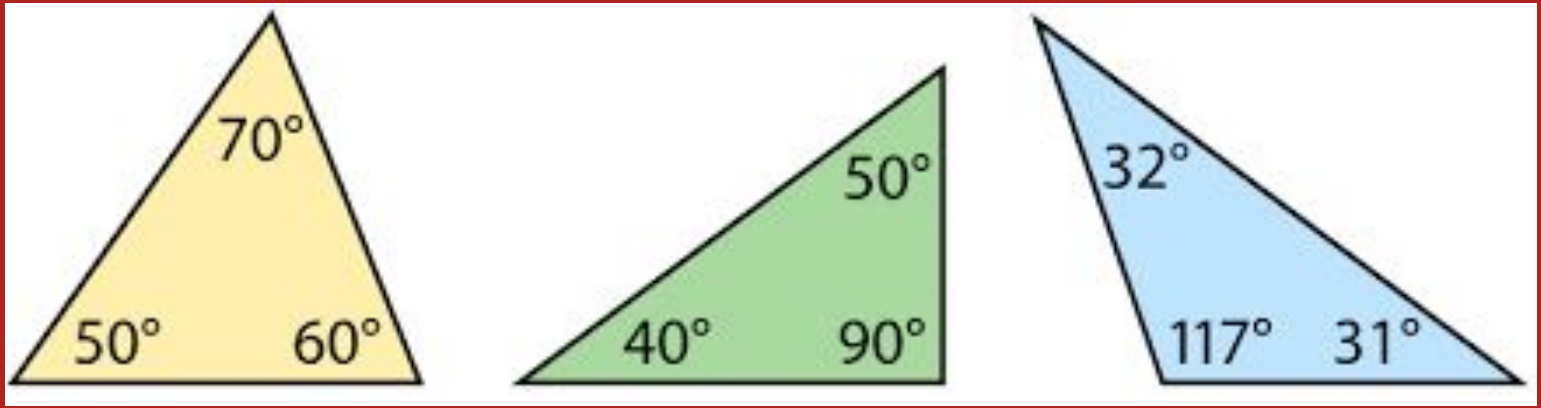


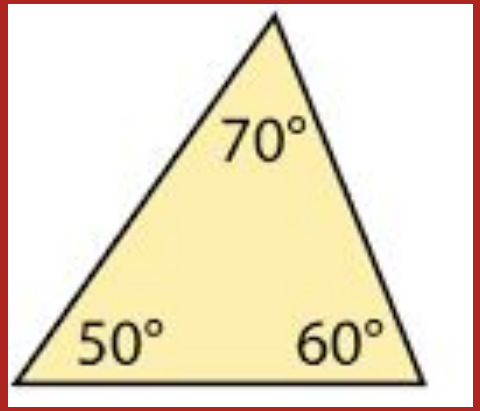
# Learning Goal

- We are learning how to find and describe interior angles, exterior angles, and opposite angles in different shapes and diagrams.

# Triangles: What Do You Notice?

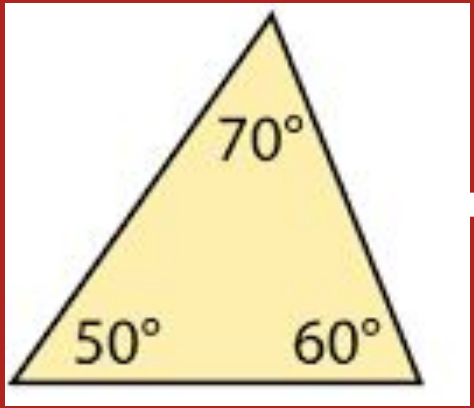


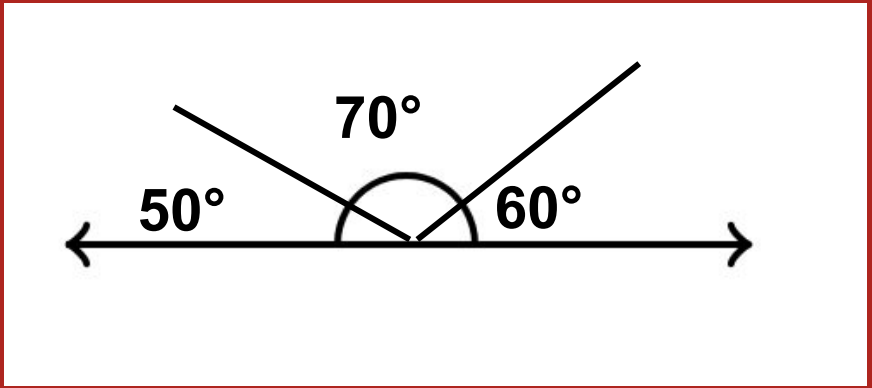
# Interior Angles



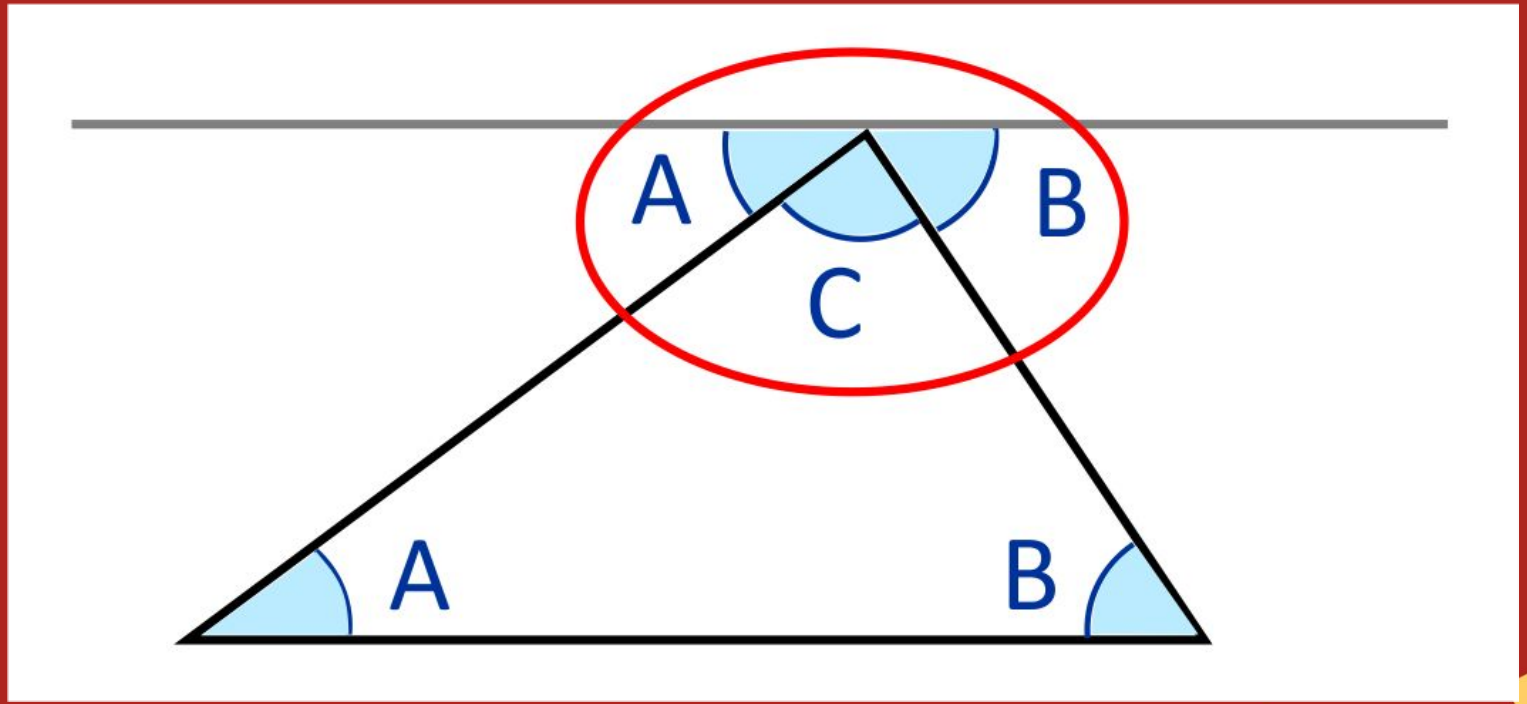

- Interior angles are the angles inside a shape.
- Every triangle has three interior angle.
- The interior angles <u>ALWAYS</u> add up to 180°.

# Why Do Interior Angles Add to 180°?

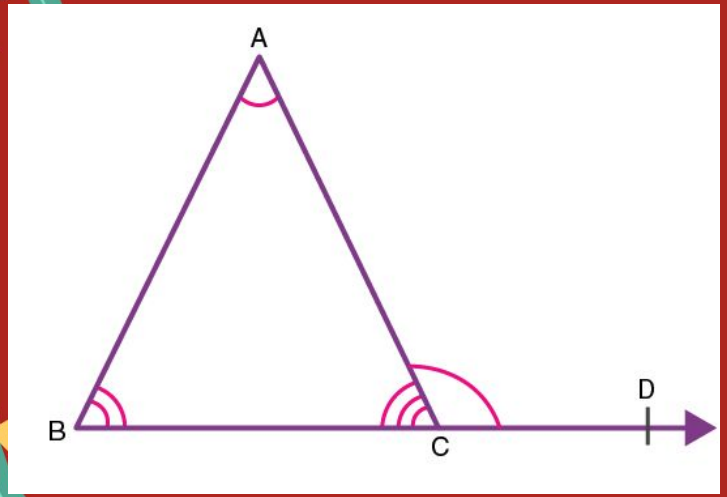


# Why Do Interior Angles Add to 180°?

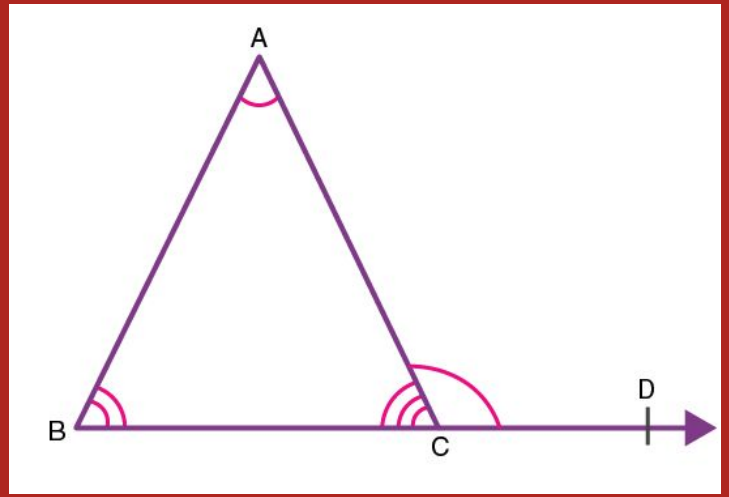


# What Are Exterior Angles?



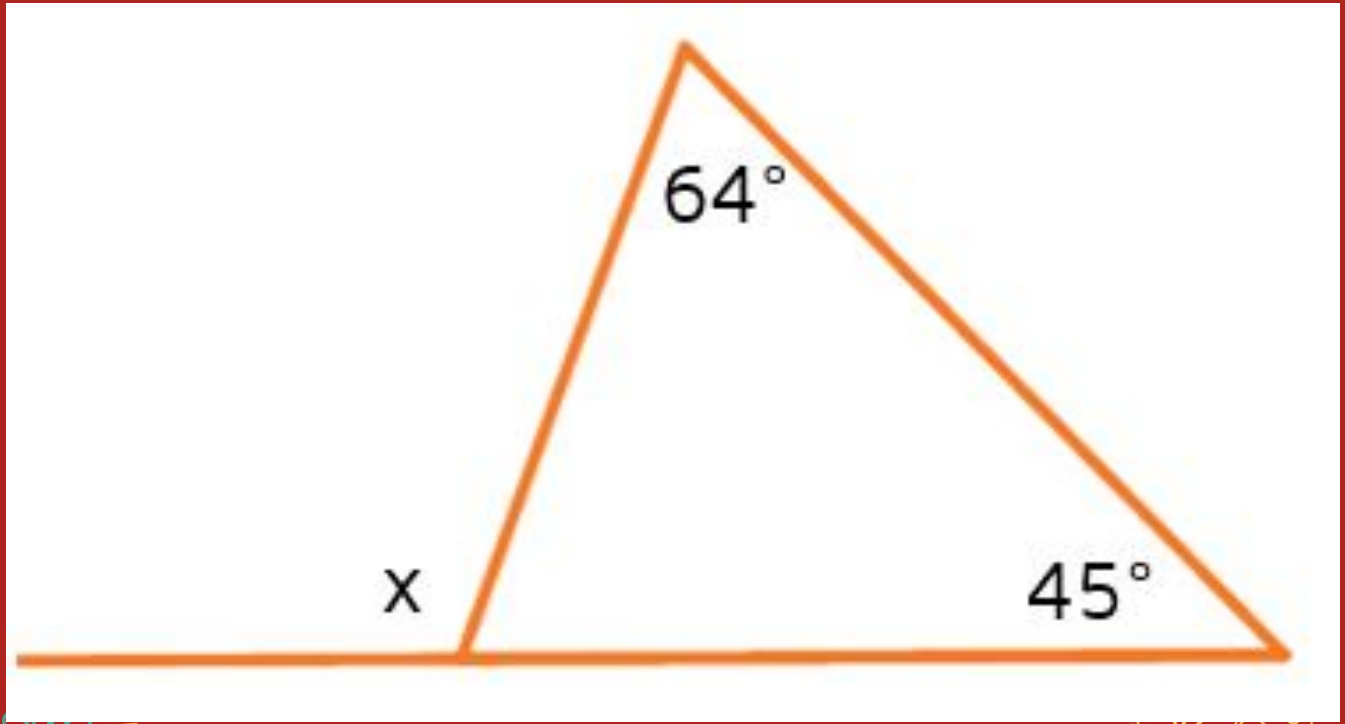

- Exterior angles are angles that are outside the shape
- They are formed by extending a side of the triangle (or shape)
- Each exterior angle is next to an interior angle.

# How Can We Use Interior / Exterior Angles?

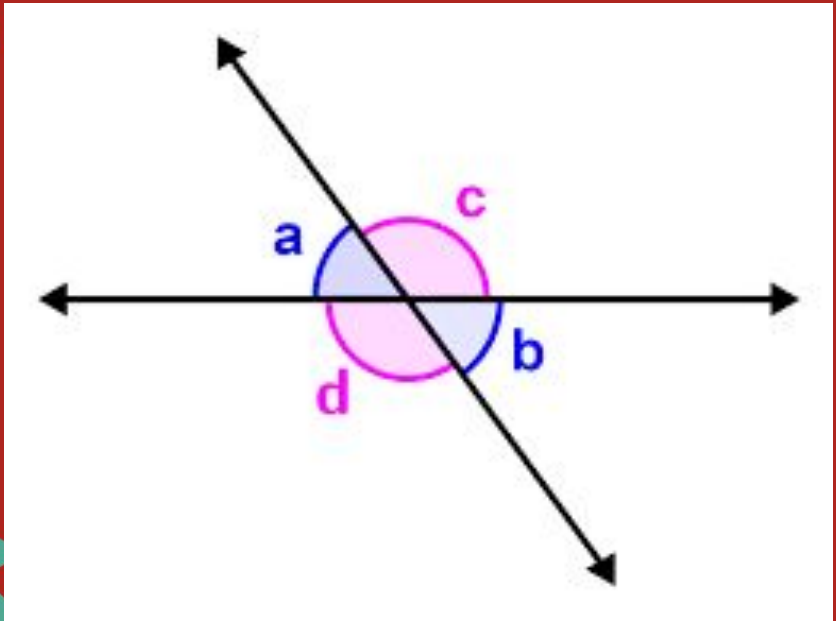



- If we look at this diagram what pops out to us? What looks familiar?
- We can see that an interior angle with an exterior angle is 180°!

# Example: Interior and Exterior Angles

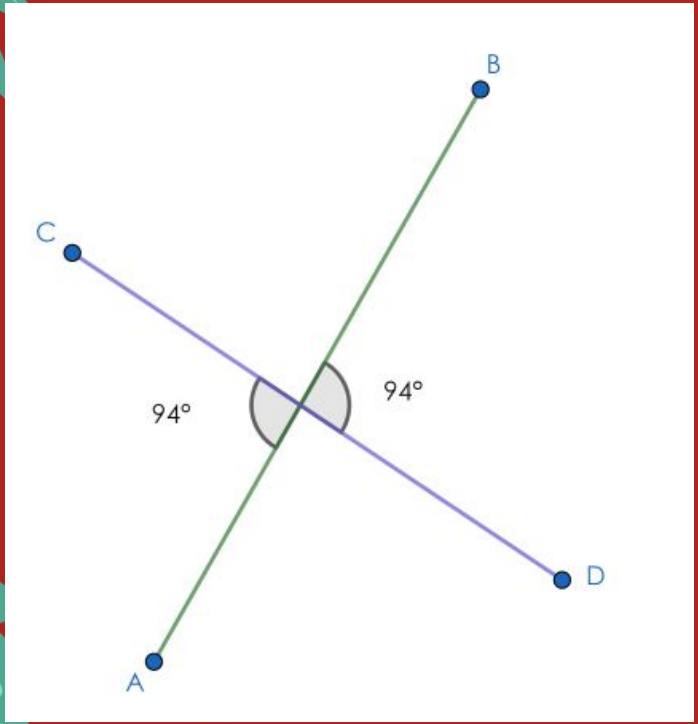


# When Two Lines Cross...




- Looking at this diagram what do we notice?

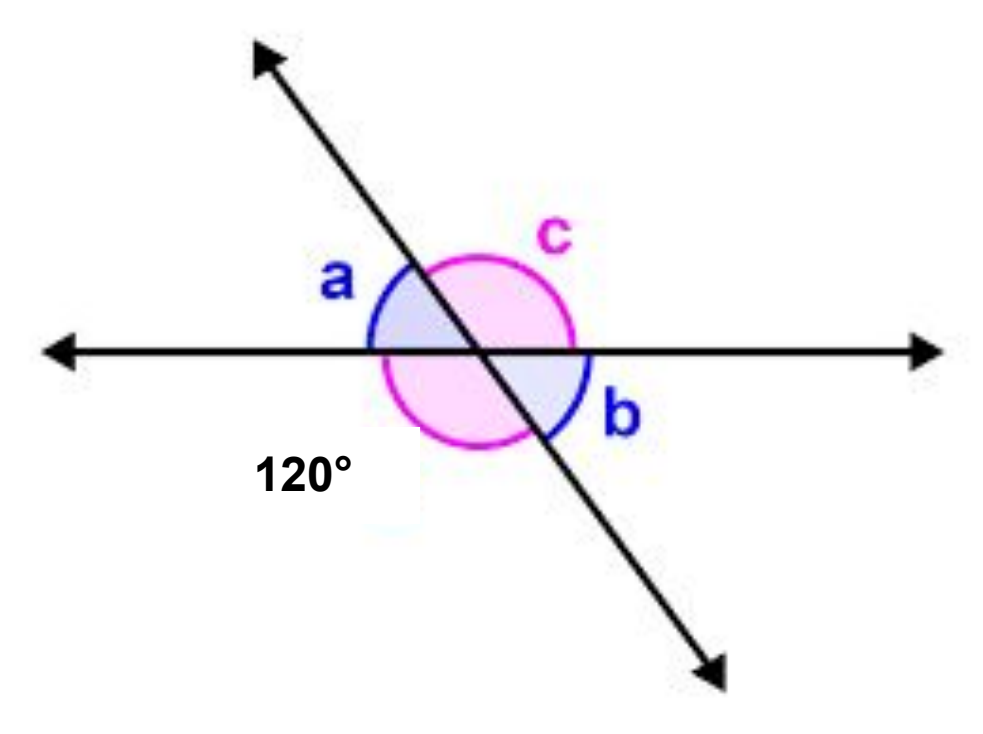
# Opposite Angles Are Equal!



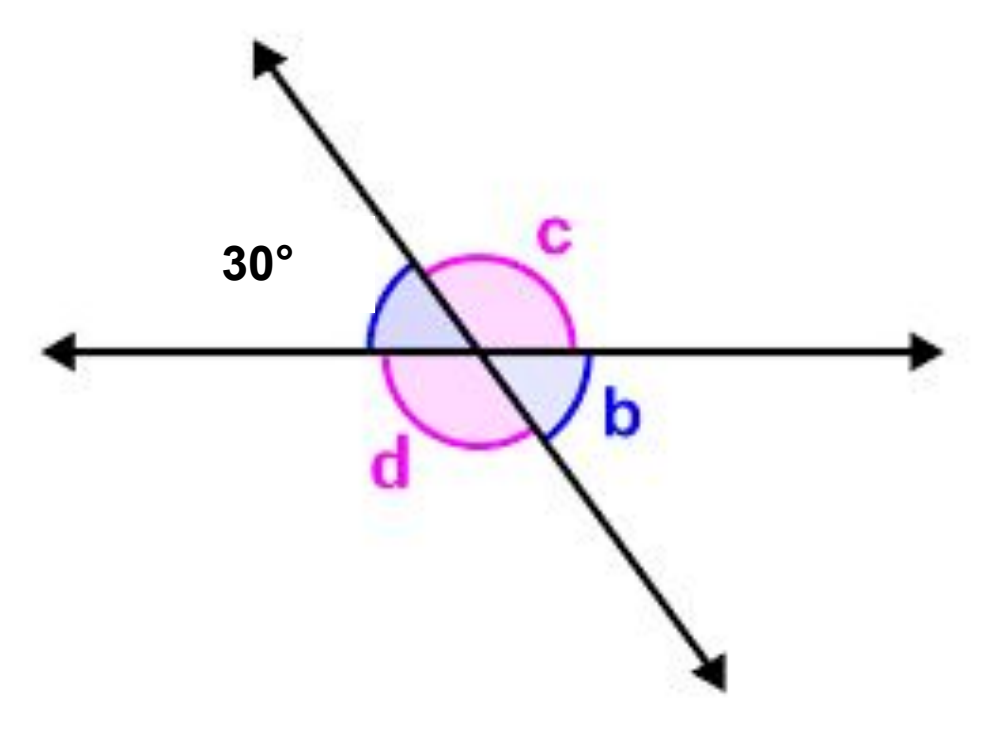

When two lines cross, they create four angles.

- Two pairs of angles share a straight line (180°).
- We can use the same rule from interior/exterior angles.
- If we know one angle we can find the other angle.

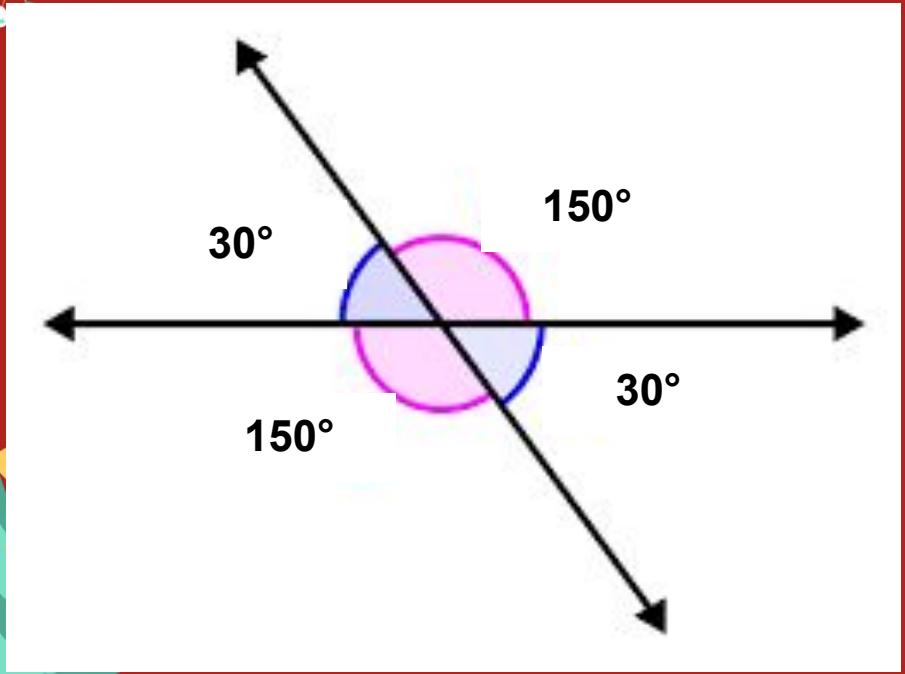
# Practice: Opposite Angles



# Practice: Opposite Angles



# Angles Around a Point




- Looking back at our examples, what do we notice about the angles?
- What number do these angles add up to?
- This means that a full turn around a point = 360°.
- All four angles created by two intersecting lines add up to 360°.

# Summary of Today's Angle Types

| Interior Angles | Exterior Angles | Opposite Angles |
|---|---|---|
| • Interior angles are inside the shape | • Exterior angles are outside the shape | • Opposite angles are formed when two lines cross |
| • In triangles, interior angles always add up to 180° | • Exterior angles are made by extending one side<br>• Interior + exterior = 180° | • Opposite angles sit directly across from each other<br>• Opposite angles are equal |
| • Example:<br>• 60° + 40° + 80° = 180° | • Exterior angles can use our straight angle rules | • A full rotation around a point is 360° |